

AGADIR AL MASSIRA INTL, Morocco

WMO: 602520

Lat: 30.325N

Lon: 9.413W

Elev: 250

StdP: 14.56

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.9	43.0	22.6	17.2	74.5	26.6	20.7	70.8	23.5	66.1	20.2	66.1	5.3	60	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.6	102.1	68.1	95.2	66.5	90.0	66.1	72.4	87.1	71.3	84.8	70.5	83.1	11.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	105.0	75.3	67.7	102.6	74.8	66.5	98.5	73.9	36.2	87.0	35.3	85.6	34.6	83.2	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	17.9	15.8	36.8	111.7	2.5	6.2	35.0	116.2	33.5	119.9	32.1	123.4	30.3	127.9
			35.1	75.0	2.9	2.1	33.1	76.5	31.4	77.8	29.8	79.0	27.7	80.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.8	58.1	60.2	64.3	66.0	68.8	72.5	75.4	76.5	73.7	71.8	65.5	60.3
	DBStd	7.81	4.38	4.83	5.60	4.90	4.74	4.53	5.70	5.43	4.25	4.99	5.56	4.15
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	709	219	149	80	36	9	0	0	0	0	3	57	156
	CDD50	6498	252	285	444	479	582	675	787	821	712	677	465	319
	CDD65	1731	6	14	59	65	125	225	322	356	262	215	72	10
	CDH74	14144	107	230	772	666	949	1564	2678	2964	1783	1634	667	130
CDH80	5543	11	47	306	287	357	536	1171	1333	605	654	223	13	
Wind	WSAvg	7.0	6.6	7.1	7.5	7.8	7.7	7.2	6.7	6.7	6.7	6.7	6.5	6.5
Precipitation	PrecAvg	10.4	1.5	1.1	1.5	0.7	0.3	0.0	0.0	0.1	0.2	0.9	1.7	2.0
	PrecMax	22.0	6.7	6.7	6.7	4.8	4.6	0.5	0.0	1.1	3.9	5.2	14.1	6.7
	PrecMin	3.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	5.7	1.8	1.6	1.7	1.1	0.9	0.1	0.0	0.3	0.8	1.3	2.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.1	85.6	93.2	97.6	102.0	104.0	113.8	110.8	101.6	98.6	91.6	81.3
		MCWB	56.0	57.1	60.2	62.1	64.4	67.3	68.6	69.3	69.2	64.1	60.9	57.8
	2%	DB	77.0	80.6	87.7	89.2	89.4	91.8	101.0	99.0	91.4	92.9	86.1	77.4
		MCWB	55.7	56.9	58.9	61.4	65.3	68.1	69.1	70.4	68.5	64.0	61.2	56.7
	5%	DB	73.2	76.5	82.7	80.4	81.0	84.3	89.9	91.3	85.7	87.6	82.0	74.1
		MCWB	54.7	56.8	58.8	61.7	65.3	67.9	69.3	70.3	68.3	63.6	60.3	56.4
	10%	DB	69.6	71.8	77.1	75.6	77.7	80.9	84.4	86.0	82.3	82.3	77.0	70.8
		MCWB	55.2	56.8	59.2	62.1	64.2	67.5	69.2	70.0	68.7	65.2	60.3	56.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.5	64.5	66.0	67.3	69.2	71.7	73.2	74.2	73.1	71.1	69.6	65.2
		MCDB	69.5	74.0	76.7	81.2	82.0	86.9	90.8	89.6	86.7	81.5	79.1	69.5
	2%	WB	60.1	62.3	63.9	65.4	67.5	70.1	71.8	72.7	71.1	69.6	66.7	63.1
		MCDB	67.8	69.8	74.8	77.3	80.3	83.4	87.1	87.5	82.4	79.5	74.2	68.5
	5%	WB	58.6	60.3	62.6	64.0	66.4	69.1	70.8	71.6	70.1	68.4	64.6	61.3
		MCDB	66.3	68.3	73.6	75.1	78.7	81.6	84.9	85.5	81.0	77.4	72.3	67.1
	10%	WB	57.1	58.9	61.2	62.9	65.2	68.0	69.9	70.7	69.2	67.3	63.0	59.5
		MCDB	65.2	67.3	71.3	73.4	76.4	79.5	83.1	83.5	79.4	75.9	71.4	66.0
Mean Daily Temperature Range	5% DB	MDBR	24.9	24.2	24.4	21.6	20.5	19.4	20.5	20.6	19.9	22.2	23.6	23.6
		MCDBR	29.3	30.6	32.6	29.8	26.4	25.8	29.6	28.2	25.1	29.9	28.6	28.0
		MCWBR	13.3	13.1	12.0	11.0	9.8	8.8	8.5	7.8	8.1	9.9	11.1	12.6
	5% WB	MCDBR	23.3	22.9	24.2	22.9	22.0	21.3	21.9	22.0	20.3	21.0	21.6	18.9
		MCWBR	13.2	12.4	12.0	10.6	9.1	8.1	7.9	7.9	8.0	9.1	11.1	10.4
Clear-Sky Solar Irradiance	taub	0.317	0.363	0.419	0.420	0.444	0.486	0.577	0.557	0.467	0.425	0.370	0.343	
	taud	2.512	2.355	2.161	2.194	2.139	2.030	1.798	1.869	2.110	2.221	2.376	2.451	
	Ebn at Noon	291	285	274	278	270	258	234	237	256	259	268	274	
	Edh at Noon	28	36	46	47	49	55	69	64	48	40	32	29	
All-Sky Solar Radiation	RadAvg	1210	1479	1874	2174	2368	2426	2387	2178	1882	1506	1228	1075	
	RadStd	68	116	76	87	118	53	61	53	90	68	72	80	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.43	N/A	N/A	N/A	-0.58	N/A	N/A	N/A	+275	+308

Nomenclature: See separate page